

Centre Number	Candidate Number	Candidate Name
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NAMIBIA SENIOR SECONDARY CERTIFICATE

CHEMISTRY ORDINARY LEVEL

6117/3

PAPER 3 Alternative to Practical

1 hour 15 minutes

Marks 40

2024

Additional materials: Non-programmable calculator
Ruler

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Candidates answer on the Question Paper in the spaces provided.
- Write your Centre Number, Candidate Number and Name in the spaces at the top of this page.
- Write in dark blue or black pen.
- You may use a soft pencil for any diagrams, graphs or rough working.
- Do not use correction fluid.
- Do not write in the margin *For Examiner's Use*.
- Answer **all** questions.
- The number of marks is given in brackets [] at the end of each question or part question.
- You may use a non-programmable calculator.

For Examiner's Use

1	
2	
3	
4	
5	
TOTAL	

Marker	
Checker	

This document consists of **9** printed pages and **3** blank pages.



Republic of Namibia
MINISTRY OF EDUCATION, ARTS AND CULTURE

- 1 A student investigated the dyes contained in different coloured inks using chromatography. Water was the solvent. Fig. 1.1 shows how the student set up the apparatus.

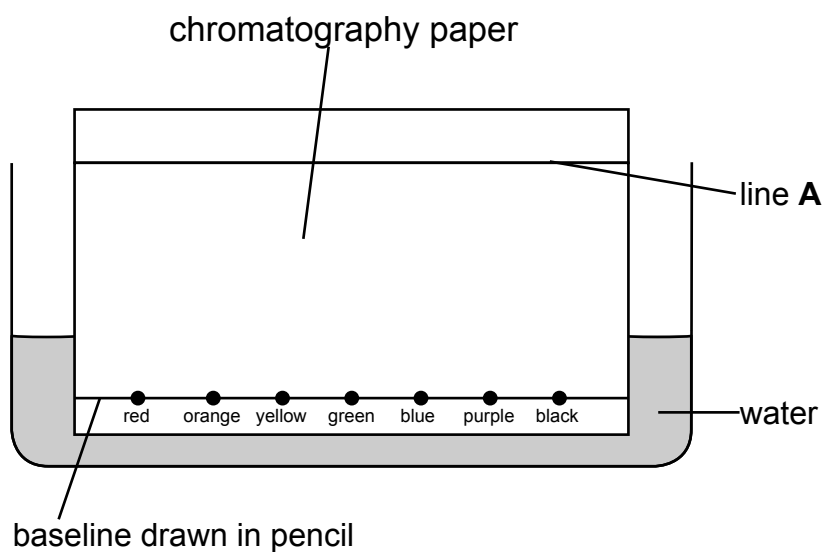


Fig. 1.1

- (a) Name line **A** from Fig. 1.1.

..... [1]

- (b) Identify an error in the way the student set up the apparatus.

..... [1]

- (c) State the reason for drawing the baseline with a pencil.

..... [1]

The student then carried out the chromatography correctly.

Fig. 1.2 shows the chromatogram.

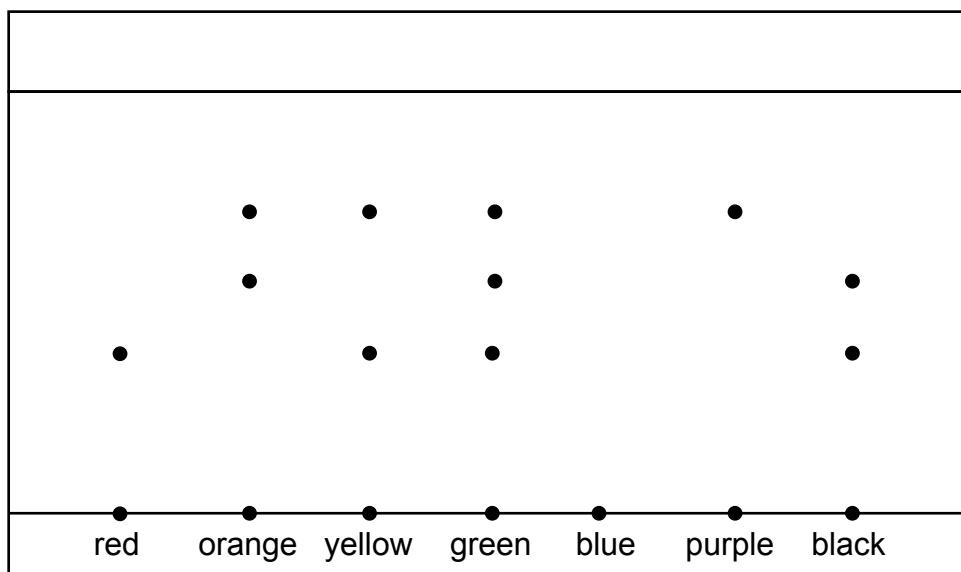


Fig. 1.2

(d) Identify the ink that contains the greatest number of dyes.

..... [1]

(e) State the **two** inks made of a single dye.

..... and [1]

(f) From the chromatogram, it is not possible to tell if the blue ink contains different dyes.

Suggest how the experiment could be changed to find out if the blue ink contains different dyes.

.....
..... [1]

[6]

- 2** A small sample of rock salt contains sodium chloride and sand.

A student obtained dry crystals of pure sodium chloride from a lump of rock salt.

These are some of the steps the student followed.

Step 1 Grind the rock salt into smaller pieces.

Step 2 Add the rock salt to water in a beaker and heat while stirring with a glass rod.

Step 3 Filter the mixture to remove the sand.

- (a) Name the **two** appropriate pieces of apparatus to be used in carrying out **step 1**.

1
2 [2]

- (b) State the reason for adding the rock salt to water in **step 2**.

.....
..... [1]

- (c) (i) Draw a diagram of the apparatus required to carry out **step 3**.

[2]

- (ii) State the term for the sand obtained in **step 3**.

..... [1]

- (d) Describe what the student must do after **step 3** to obtain dry crystals of pure sodium chloride.

.....
.....
..... [2]

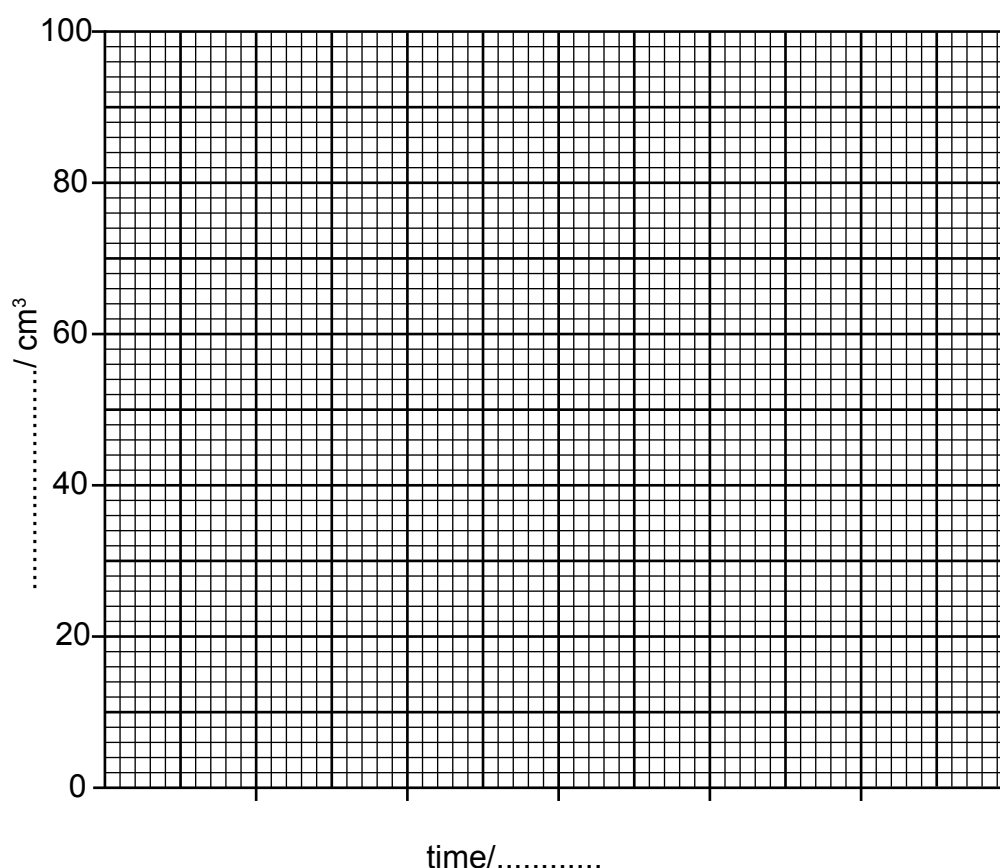
[8]

- 3 When potassium chlorate(V) is heated, it decomposes and oxygen is evolved. A student heats a sample of potassium chlorate(V) crystals for 120 seconds. The volume of oxygen collected is measured in a gas syringe for every 20 seconds.
- (a) Complete the table by filling in the column headings and all the values for time.

time/.....	 /cm ³
0	0
	30
	50
	66
	78
	86
	86

[3]

- (b) Plot the results for the experiment on the grid and draw a smooth curve. Complete your axes.



[3]

- (c) From your graph, predict the volume of oxygen produced after 45 seconds. Show clearly on the grid how you worked out your answer.

volume = cm³ [2]

(d) On the grid, sketch the graph you would expect if the experiment was repeated using potassium chlorate(V) powder instead of crystals.

[2]

(e) Explain why the final two readings recorded in the table are the same.

.....

.....

[1]

(f) Describe the chemical test for oxygen.

test

result

[2]

[13]

- 4 Two solids, **N** and **P**, were analysed.

Tests were done on each solid.

Tests on solid N.

tests on solid N	observations
Solid N was dissolved in distilled water to produce solution N . The solution was divided into three equal portions in three boiling tubes.	
test 1 Aqueous sodium hydroxide was added slowly until in excess to the first portion of solution N .	white precipitate formed, the precipitate dissolved in excess forming a colourless solution
test 2 Aqueous ammonia was added slowly until in excess to the second portion of solution N .	white precipitate formed, the precipitate does not dissolve in excess
test 3 The third portion of solution N was acidified with dilute nitric acid, then aqueous silver nitrate was added.	yellow precipitate formed

- (a) Identify solid **N**.

.....
 [2]

Tests on solid P.

Solid **P** was potassium sulfate.

- (b) (i) Describe the appearance of solid **P**.

.....
 [1]

- (ii) State the observation made when a flame test is done on solid **P**.

.....
 [1]

Solid **P** was dissolved in distilled water to produce solution **P**.
Solution **P** was divided into two equal portions in two test-tubes.

- (iii) About 1 cm depth of dilute hydrochloric acid was added to the first portion of solution **P**.

Describe the expected observation.

.....
.....

[1]

- (iv) About 1 cm depth of dilute nitric acid and a few drops of aqueous barium nitrate were added to the second portion of solution **P**.

Describe the expected observation.

.....
.....

[1]

[6]

- 5 Many multi-purpose cleaning products contain aqueous ammonia. Aqueous ammonia is an alkali that reacts with dilute acids.

You are provided with:

- an aqueous solution of two multi-purpose cleaning products
- dilute hydrochloric acid of known concentration
- common laboratory apparatus.

Plan an experiment to investigate which of the two multi-purpose cleaning products contains the most concentrated aqueous ammonia.

Your answer should include

- a brief explanation of the method you will use, including the readings you would take.
- how your results will be used to reach a conclusion.
- one key variable that you would control to ensure a fair test.

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[7]

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